Chapter 5.1 — Formula Sheet

Joint, Marginal & Conditional Probability Distributions

Companion quick-reference to your Notebook 5 syllabus and your *20 Worked Examples* set (Example labels 1–20 cross-referenced below)

A. Joint Distributions & Marginals

Marginals by Summation / Integration [Ex 1–3]

Discrete:

$$f_X(x) = \sum_y f_{X,Y}(x,y), \qquad f_Y(y) = \sum_x f_{X,Y}(x,y)$$

Continuous:

$$f_X(x) = \int f_{X,Y}(x,y)\,dy, \qquad f_Y(y) = \int f_{X,Y}(x,y)\,dx$$

Watch the limits: on a non-rectangular support (e.g. a triangle), the integration range for y depends on x — the single most common error source.

Normalizing Constants [Ex 4]

For $f(x,y) = c \cdot g(x,y)$, solve

$$\iint_{\text{support}} c \cdot g(x,y)\,dx\,dy = 1 \quad (\text{or } \textstyle\sum\sum = 1 \text{ discrete})$$

Never assume $c = 1$ — always integrate/sum the *un-normalized* shape over its actual support first.

The Missing-Cell Trick [Ex 5]

In an $m \times n$ discrete table, only $mn - 1$ cells are free — the last is forced:

$$\text{(unknown cell)} = 1 - \text{(sum of all other cells)}$$

B. The Independence Criterion

Core Test [Ex 6–7]

$$X \perp Y \iff f_{X,Y}(x,y) = f_X(x)\,f_Y(y) \quad \text{for } \textit{every } (x,y)$$

One mismatched point/cell is enough to convict dependence — it's an all-points promise, not a majority vote. Discrete shortcut: independence $\iff$ every row of the table is a scalar multiple of every other row.

Two Traps [Ex 9–10]

Trap 1 (non-rectangular support): a constant/"uniform-looking" density on a triangle is *still dependent* — the support itself couples X and Y. **Trap 2** (rectangle isn't enough): even on a perfect rectangle with both marginals *individually* uniform, the joint can fail to factor (e.g. $f = \frac{1}{4}(1+xy)$) — always check the joint density itself, not just the marginal shapes.

Factoring Shortcut [Ex 8]

If $f(x,y) = g(x)h(y)$ *and* the support is a rectangle (no x-y coupling in the limits), independence holds on sight — the marginals are just g, h rescaled to integrate to 1.

C. Conditional Distributions

Core Definition [Ex 11–12]

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}, \qquad f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

Treat the conditioning variable as a fixed constant. **Built-in check:** the result must integrate/sum to exactly 1 over the free variable, for *any* fixed value of the conditioning variable.

Recognizing a Named Conditional Law [Ex 13]

Uniform joint density on a triangle $\{0 < y < M(x)\} \Rightarrow$

$$Y \mid X{=}x \sim \text{Uniform}\big(0,\, M(x)\big)$$

Skip the integral: quote $\mathbb{E}[Y \mid X{=}x] = M(x)/2$ and $\mathbb{P}(Y > t \mid X{=}x) = \frac{M(x)-t}{M(x)}$ directly.

Conditional Expectation & Tail Probabilities [Ex 14–15]

$$\mathbb{E}[Y \mid X = x] = \int y\, f_{Y|X}(y \mid x)\,dy \quad \text{or} \quad \sum_y y\, f_{Y|X}(y \mid x)$$

Once the conditional law is identified by name, probability questions about it are read off directly (a "ruler measurement"), not re-derived from scratch.

D. Capstones — Traps & Applications

Independent ⇒ Uncorrelated; Converse False [Ex 16]

$$\text{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Independence always forces $\text{Cov} = 0$. The converse fails: $\text{Cov}(X,Y) = 0$ *never* proves independence — covariance only detects *linear* co-movement (classic case: $Y = X^2$ with X symmetric about 0).

Conditioning in Trading Applications [Ex 17]

For correlated payoffs, $\mathbb{E}[Y \mid X{=}x]$ can differ sharply from $\mathbb{E}[Y]$ — restrict to the row/column, renormalize by $f_X(x)$, take the weighted average. Correlated order flow means conditioning on one asset's outcome genuinely moves your estimate of another's.

The Full Pipeline [Ex 18–20]

One connected skill, run in order on any new density:

(1) Normalize: find c → (2) Marginalize: f_X, f_Y

→ (3) Test independence: one mismatch convicts

→ (4) Condition: $f_{Y|X} = \frac{f_{X,Y}}{f_X}$

→ (5) Integrate for a probability

Every "hard" joint-distribution question is this five-step chain, timed.

One-tool-tree reminder: a joint density is the full picture; a marginal is one shadow of it (sum/integrate the other variable away); independence is the strongest possible statement about that picture (it factors, everywhere); and a conditional distribution is what's left after you've already been told part of the picture. Normalize, marginalize, test, condition — the same four verbs, on every density this chapter throws at you.